Task-19:- Technical Documentation-(SMTP, AWS SES, and AWS WorkMail)

1. Introduction

1.1 Purpose

The purpose of this document is to define the implementation of a monitoring and alerting system using Prometheus and Grafana integrated with multiple email delivery mechanisms.

The system monitors system-level metrics such as CPU and memory usage and sends alert notifications using different SMTP configurations including internal SMTP (Postfix), AWS SES, and AWS WorkMail.

1.2 Scope

In Scope

- Installation of Node Exporter
- Installation and configuration of Prometheus
- Installation and configuration of Grafana
- Integration of Prometheus with Grafana
- Importing Node Exporter dashboard
- Creating alert rules (CPU, Memory)
- Configuring email alerts using:
  - Internal SMTP (Postfix)
  - AWS SES SMTP
  - AWS WorkMail SMTP
- Testing alert delivery
- Validating email notifications

Out of Scope

- Kubernetes monitoring
- CI/CD pipelines
- Auto Scaling
- Third-party integrations (Slack, PagerDuty)

1.3 Definitions

Term	Meaning
Prometheus	Monitoring tool
Node Exporter	System metrics exporter
Grafana	Visualization and alerting tool
SMTP	Email sending protocol
AWS SES	Cloud email service
AWS WorkMail	Managed email service

1.4 References

- Prometheus Documentation
- Grafana Documentation
- AWS SES Documentation
- AWS WorkMail Documentation

1.5 Overview

The system collects metrics from the server using Node Exporter, stores them in Prometheus, and visualizes them in Grafana. Alert rules are created in Grafana and notifications are sent through different email systems to ensure reliability.

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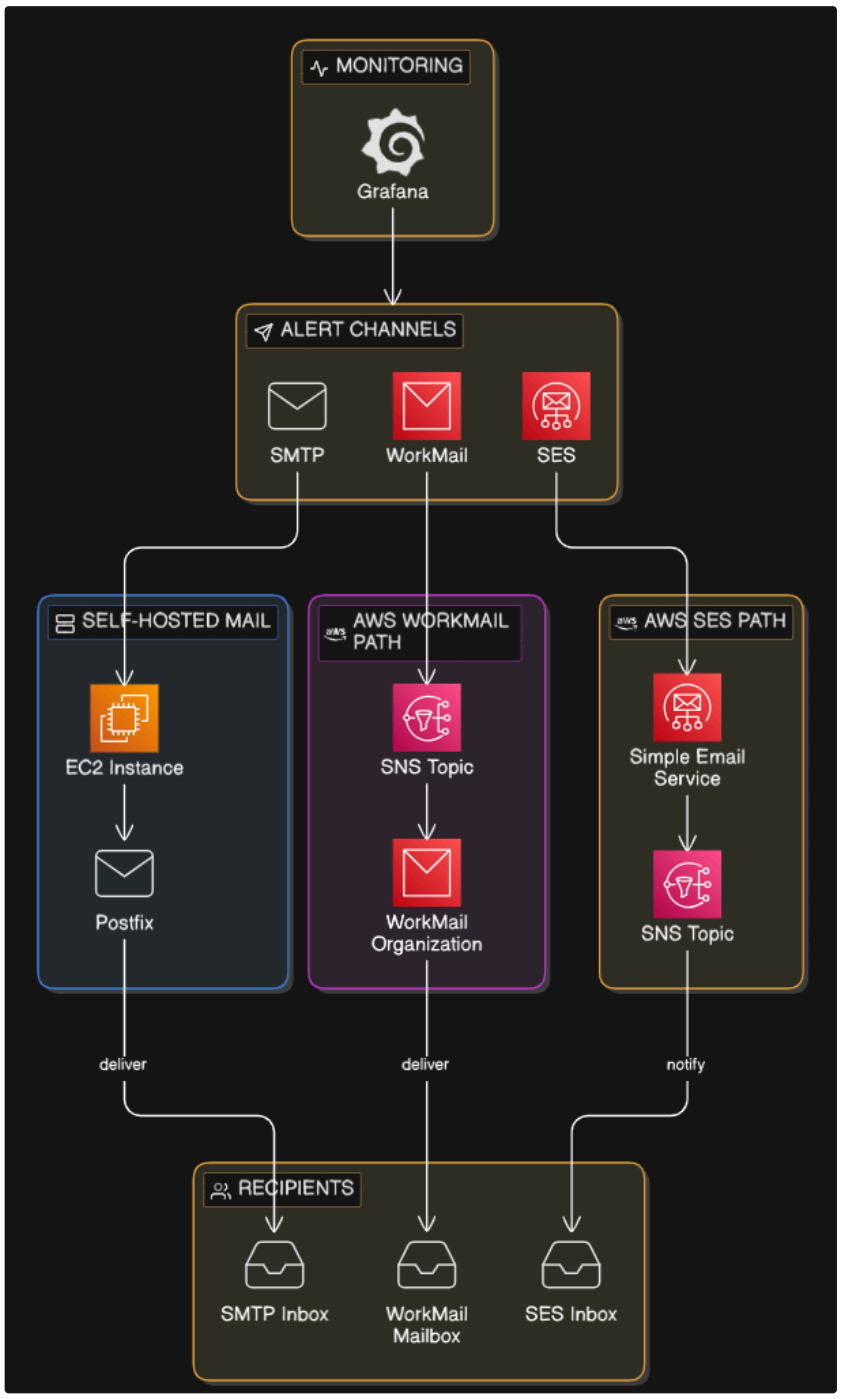
2. System Architecture



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3. System Design

Components

- Node Exporter
- Prometheus
- Grafana

- SMTP Systems (Postfix, AWS SES, AWS WorkMail)

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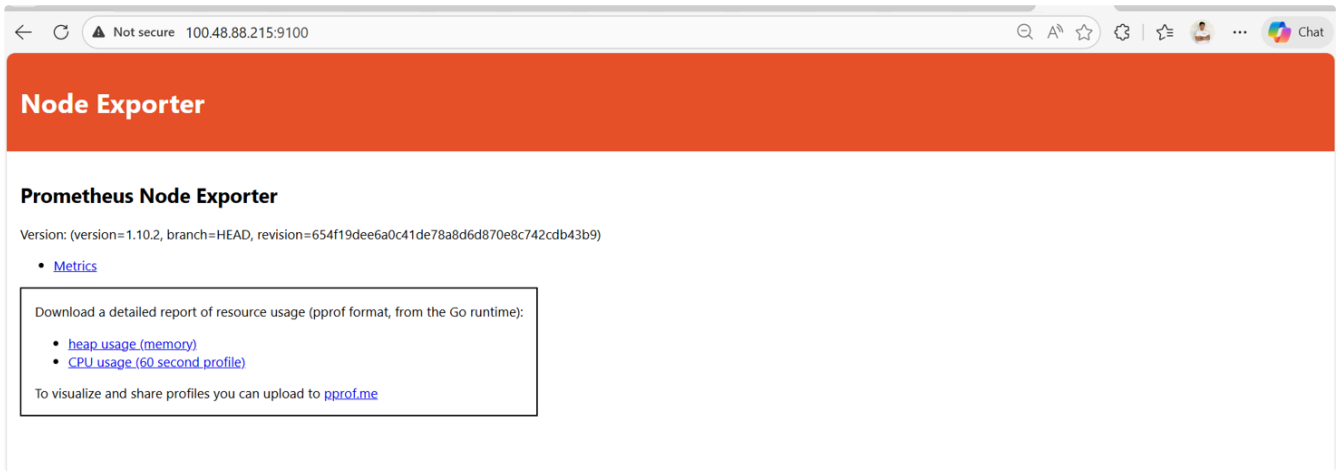
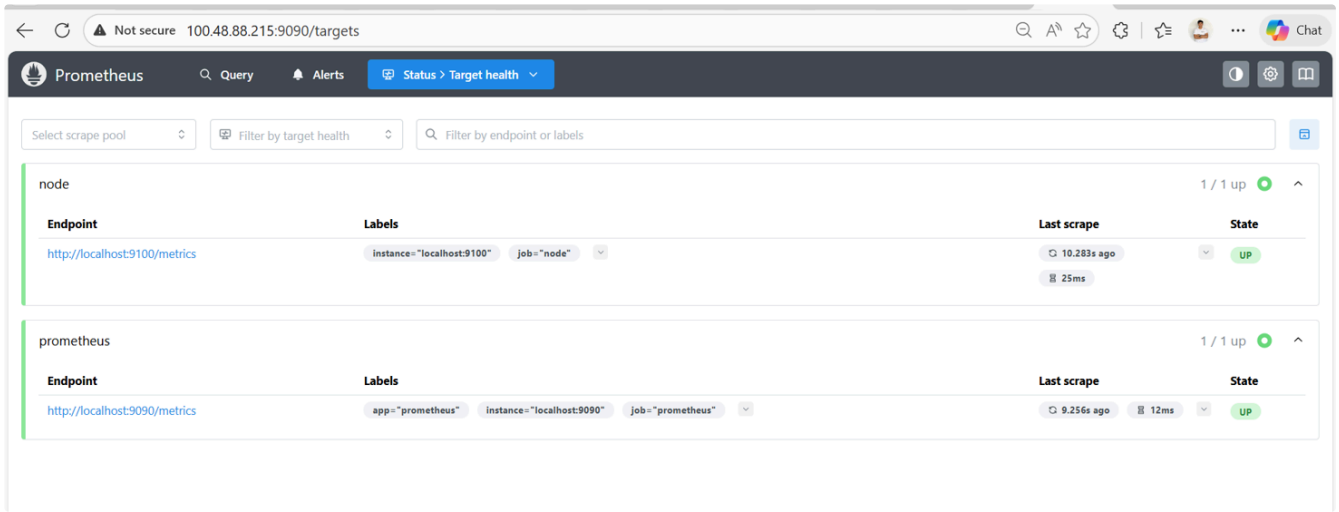
4. Implementation

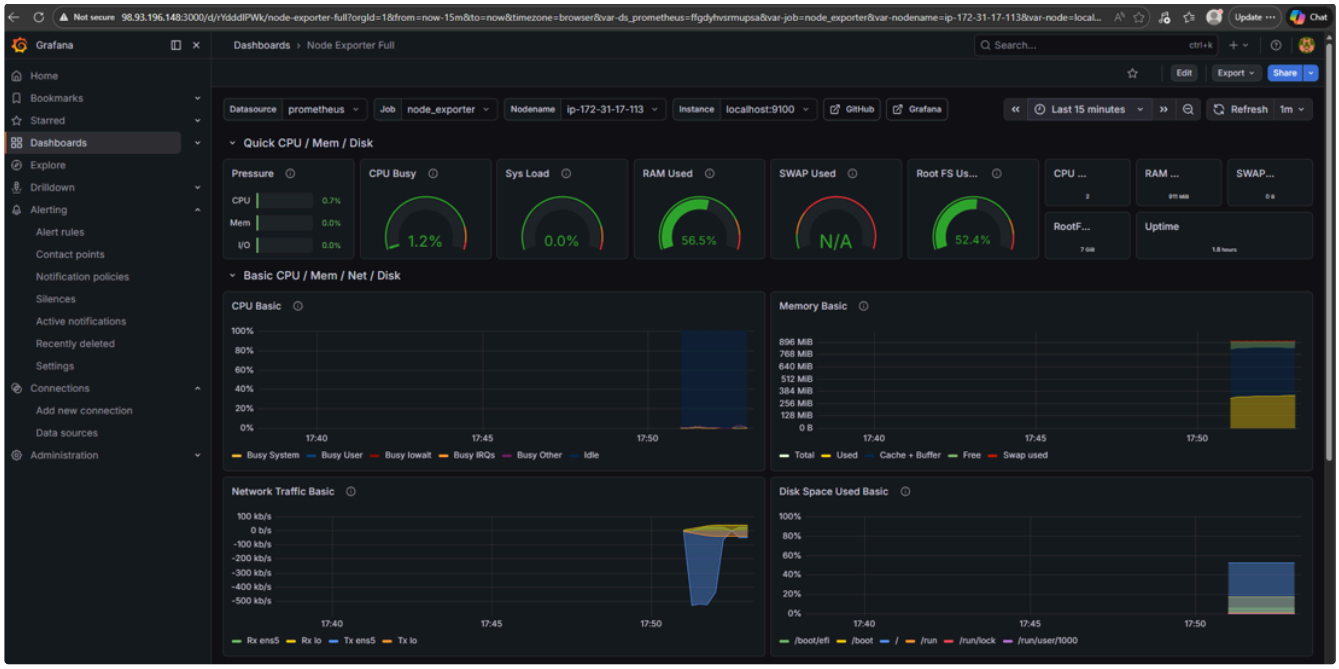
4.1 Node Exporter Setup

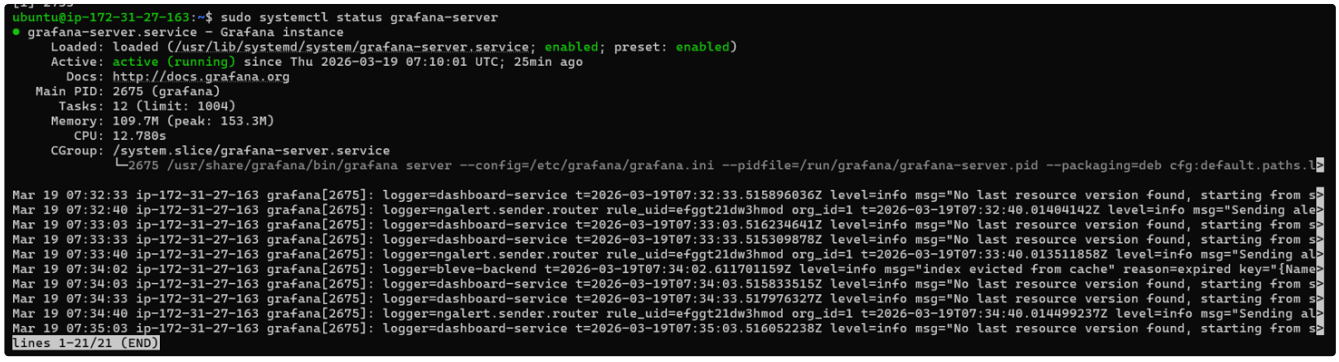


4.2 Prometheus Setup



4.3 Grafana Setup

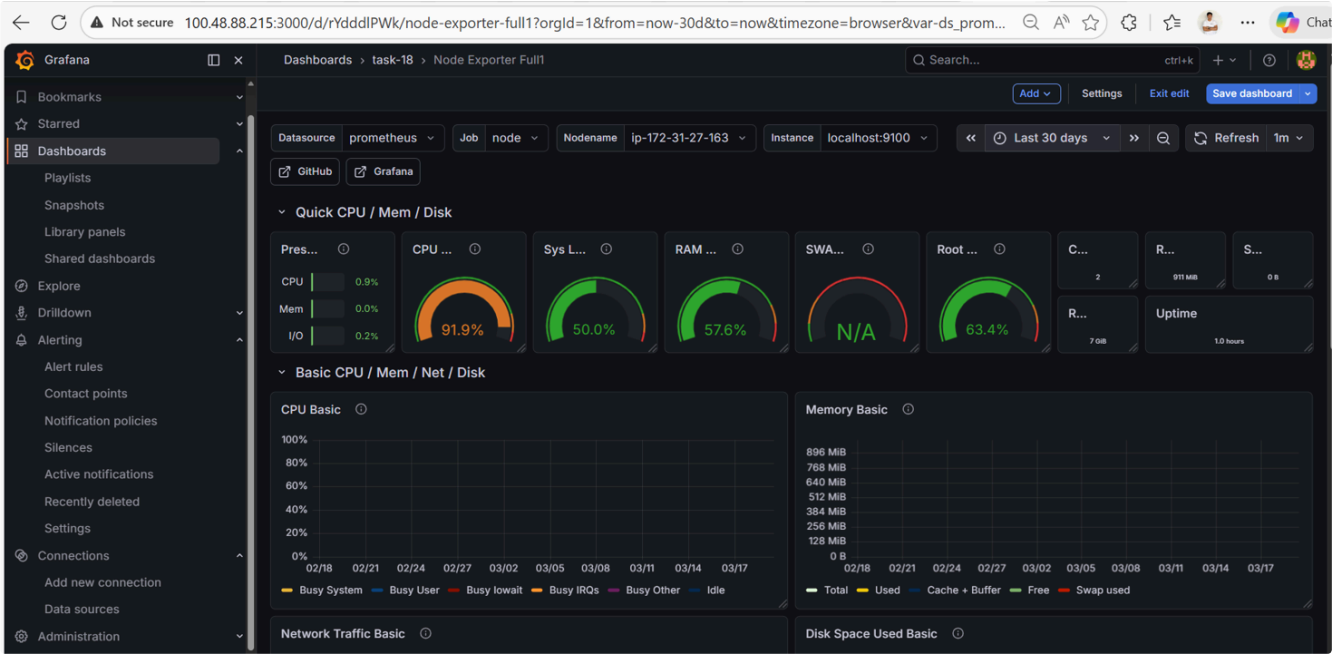




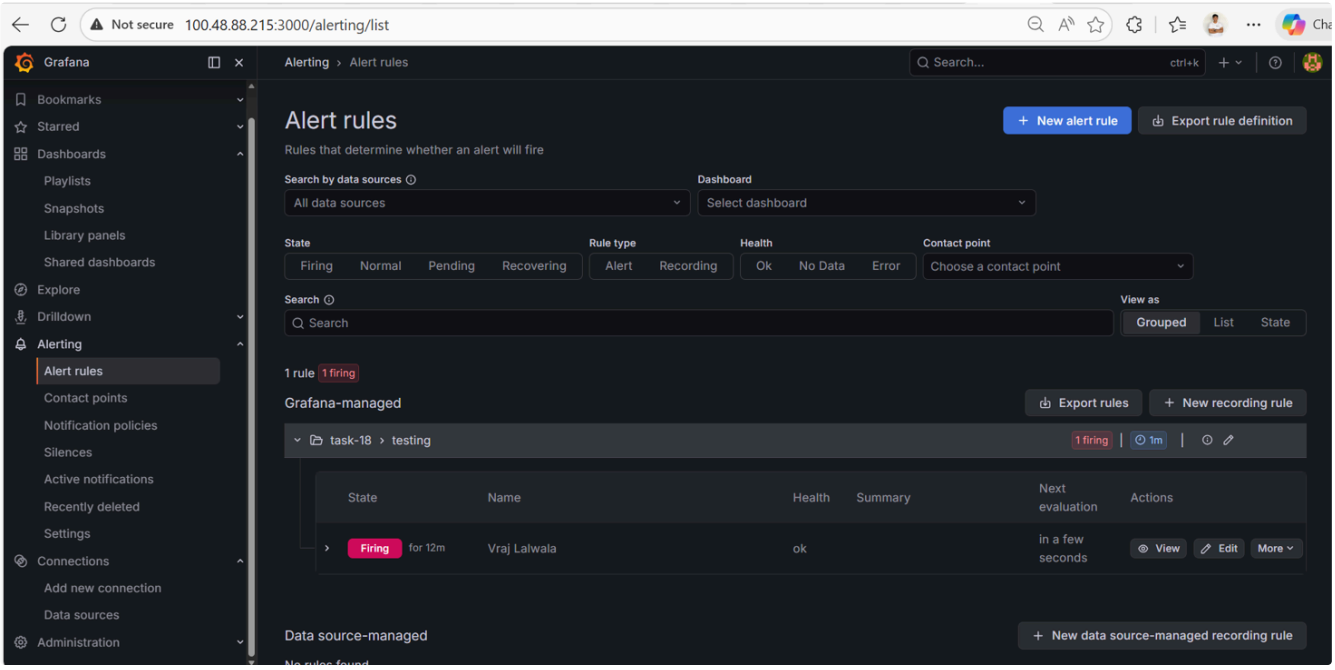
4.4 Alert Configuration

CPU Alert

```
100 - (avg by(instance)(rate(node_cpu_seconds_total{mode="idle"}[1m])) * 100)
```



Memory Alert:-



4.5 Email Notification Configuration

Method 1: Internal SMTP (Postfix + SES Relay)

Postfix Setup:-

```
ubuntu@ip-172-31-16-96:~$ sudo apt install postfix mailutils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gsasl-common guile-3.0-libs libgc1 libgsasl18 libgssglue1 libidn12 libltdl7 libmailutils9t64 libmysqlclient21 libnsl2 libntlm0
  libpq5 mailutils-common mysql-common ssl-cert
Suggested packages:
  mailutils-mh mailutils-doc postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver postfix-mysql postfix-pcre
  postfix-pgsql postfix-sqlite procmail sasl2-bin | dovecot-common
The following NEW packages will be installed:
  gsasl-common guile-3.0-libs libgc1 libgsasl18 libgssglue1 libidn12 libltdl7 libmailutils9t64 libmysqlclient21 libnsl2 libntlm0
  libpq5 mailutils mailutils-common mysql-common postfix ssl-cert
0 upgraded, 17 newly installed, 0 to remove and 114 not upgraded.
Need to get 11.7 MB of archives.
After this operation, 72.6 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libnsl2 amd64 1.3.0-3build3 [41.4 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 postfix amd64 3.8.6-1build2 [1254 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 gsasl-common all 2.2.1-1willsync1build2 [5056 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libgc1 amd64 1:8.2.6-1build1 [90.3 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 guile-3.0-libs amd64 3.0.9-1build2 [7630 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libgssglue1 amd64 0.9-1build1 [20.7 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libidn12 amd64 1.42-1build1 [55.9 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libntlm0 amd64 1.7-1build1 [19.4 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libgsasl18 amd64 2.2.1-1willsync1build2 [72.8 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libltdl7 amd64 2.4.7-7build1 [40.3 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 mailutils-common all 1:3.17-1.1build3 [389 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mysql-common all 5.8+1.1.0build1 [6746 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmysqlclient21 amd64 8.0.45-0ubuntu0.24.04.1 [1255 k
```

```
Package configuration

                    Postfix Configuration
 Please select the mail server configuration type that best meets your needs.

  No configuration:
   Should be chosen to leave the current configuration unchanged.
  Internet site:
   Mail is sent and received directly using SMTP.
  Internet with smarthost:
   Mail is received directly using SMTP or by running a utility such
   as fetchmail. Outgoing mail is sent using a smarthost.
  Satellite system:
   All mail is sent to another machine, called a 'smarthost', for
   delivery.
  Local only:
   The only delivered mail is the mail for local users. There is no
   network.

 General mail configuration type:

                          No configuration
                          Internet Site
                          Internet with smarthost
                          Satellite system
                          Local only

                    <Ok>                    <Cancel>
```

Grafana Configuration:

```
[smtp]
enabled = true
host = localhost:25
from_address = XXXXXXXXXXX@.awsapps.com
from_name = Grafana Alerts
```

Verification:

Alert Proof:-

6:35

98

[FIRING:1] High Alert TestData

Inbox

S

me 6:35 PM

to me

Grafana

TestData › High Alert

🔥 **1 firing instances**

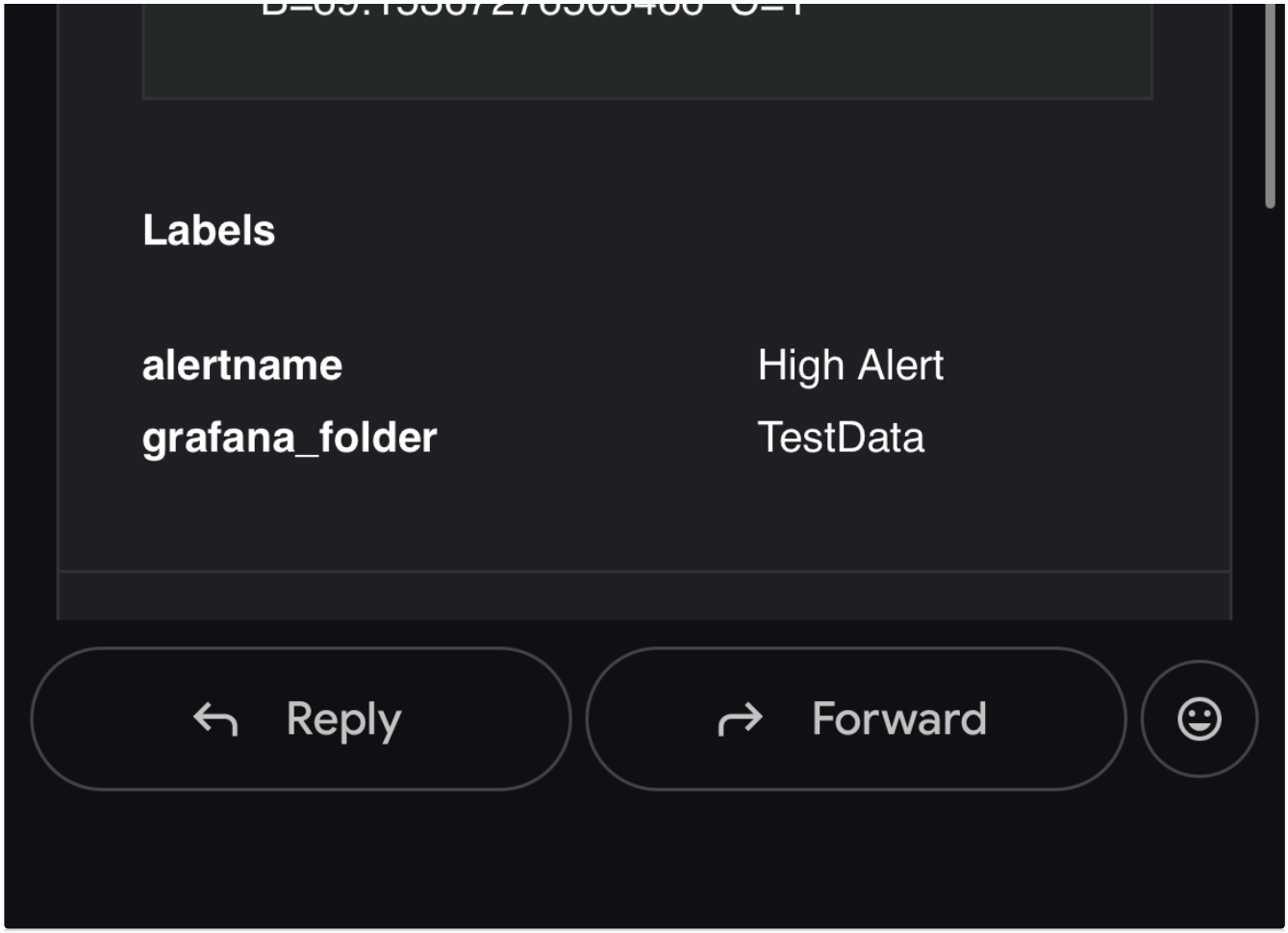
Firing

High Alert

View alert

Values

B=69.15367276503466, C=1

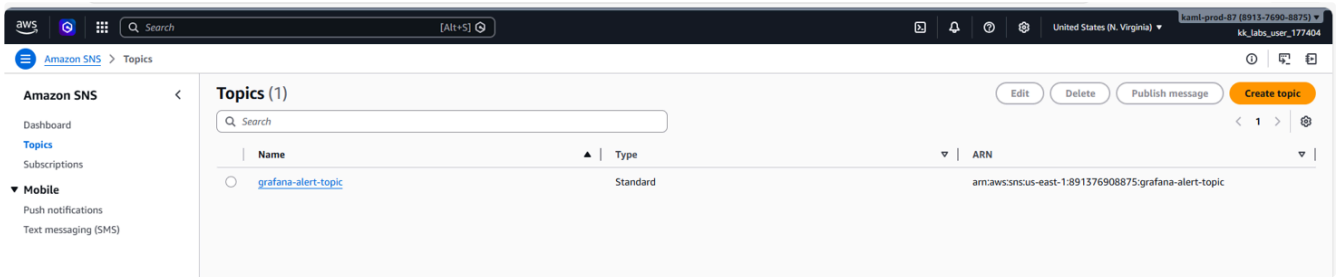


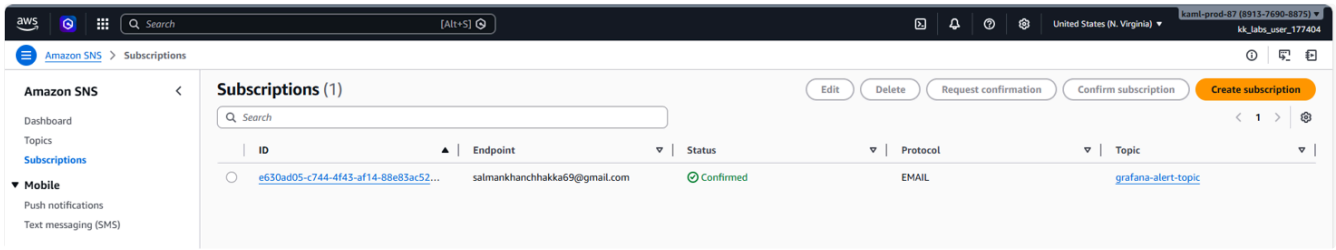
Method 2: AWS SES (Direct SMTP)

Grafana Configuration:

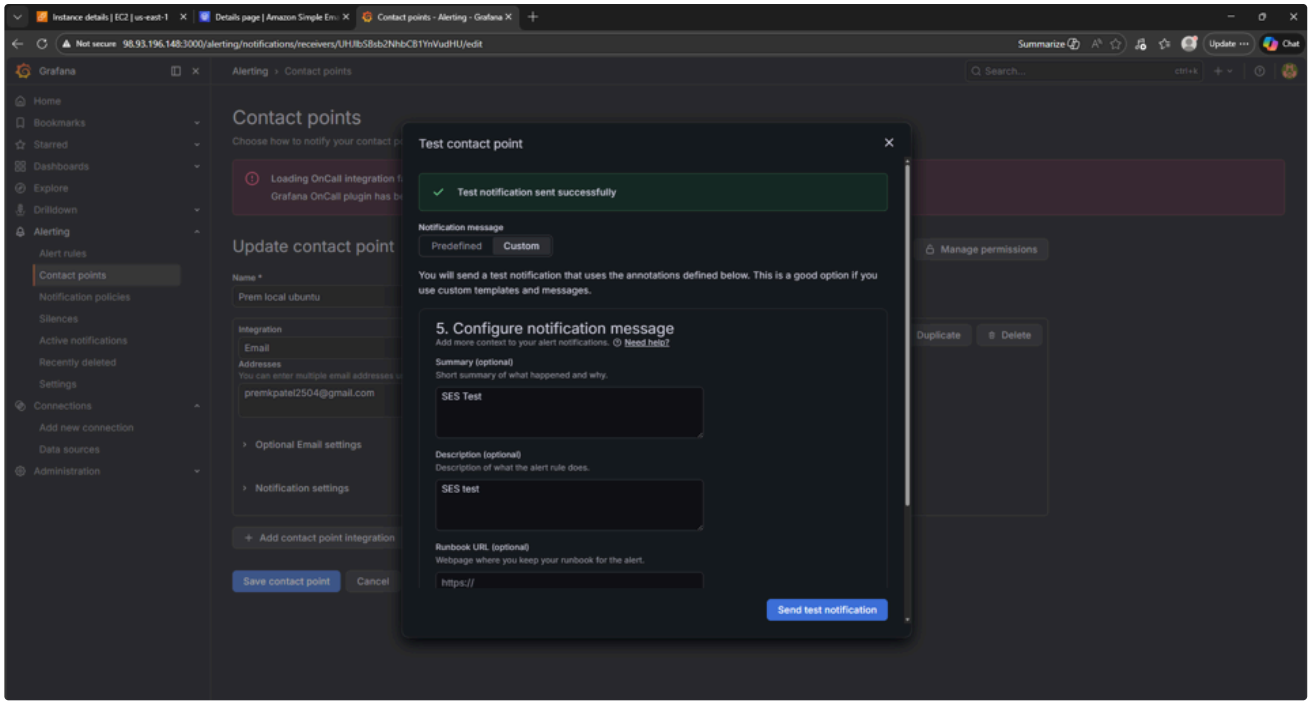
```
[smtp]
enabled = true
host = email-smtp.us-east-1.amazonaws.com:587
user = <SMTP_USERNAME>
password = <SMTP_PASSWORD>
skip_verify = true

```

```
[smtp]
enabled = true
host = localhost:25
;user =
# If the password contains # or ; you have to
;password =
;cert_file =
;key_file =
;skip_verify = false
from_address = salmankhanchhakka69@gmail.com
from_name = Grafana Alerts
```

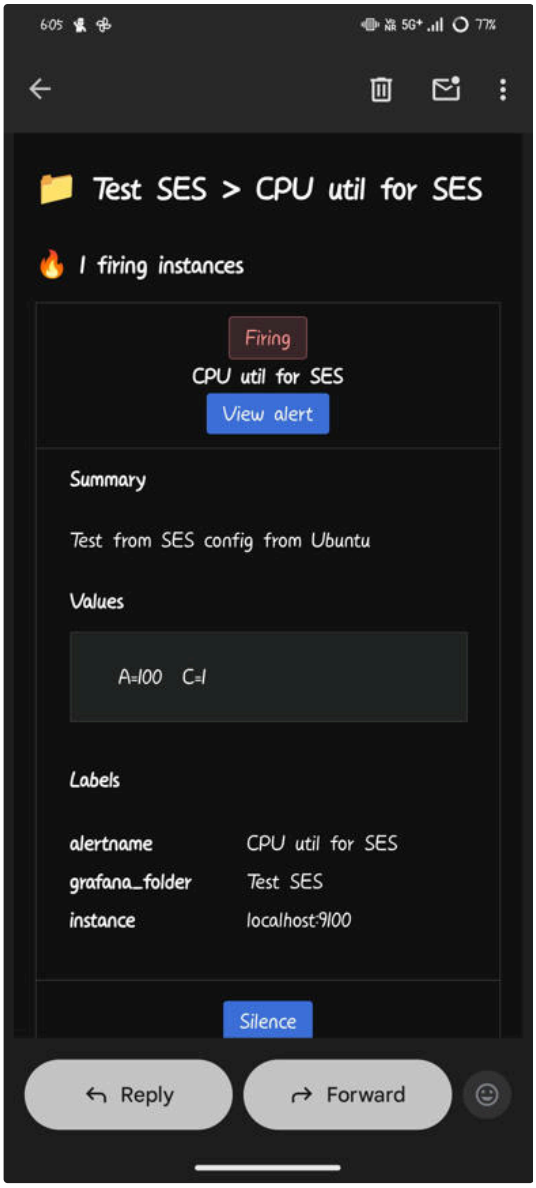


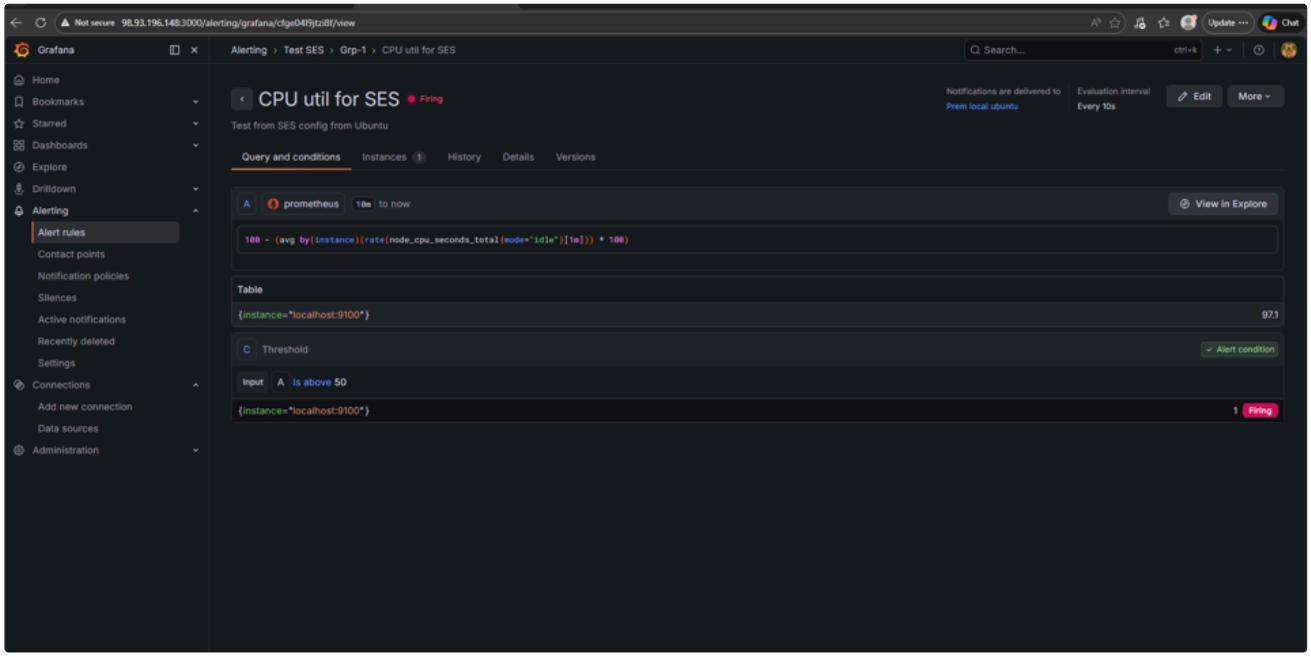


Verification:


Contact points - Alerting - Grafana
Not secure 98.93.196.148:3000/alerting/notifications/receivers/UHJlbS8sb2NhbC81YnVudHU/edit
Grafana
Alerting › Contact points
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Alert rules
Contact points
Notification policies
Silences
Active notifications
Recently deleted
Settings
Connections
Add new connection
Data sources
Administration
Contact points
Update contact point
Name *
Prem local ubuntu
Integration
Email
Addresses
premkpatel2504@gmail.com
Optional Email settings
Notification settings
Add contact point integration
Save contact point
Cancel
Manage permissions
Duplicate
Delete
Test contact point
Test notification sent successfully
Notification message
Predefined
Custom
You will send a test notification that uses the annotations defined below. This is a good option if you use custom templates and messages.
5. Configure notification message
Add more context to your alert notifications.
Need help?
Summary (optional)
Short summary of what happened and why.
SES Test
Description (optional)
Description of what the alert rule does.
SES test
Runbook URL (optional)
Webpage where you keep your runbook for the alert.
https://
Send test notification

Alert Proof:

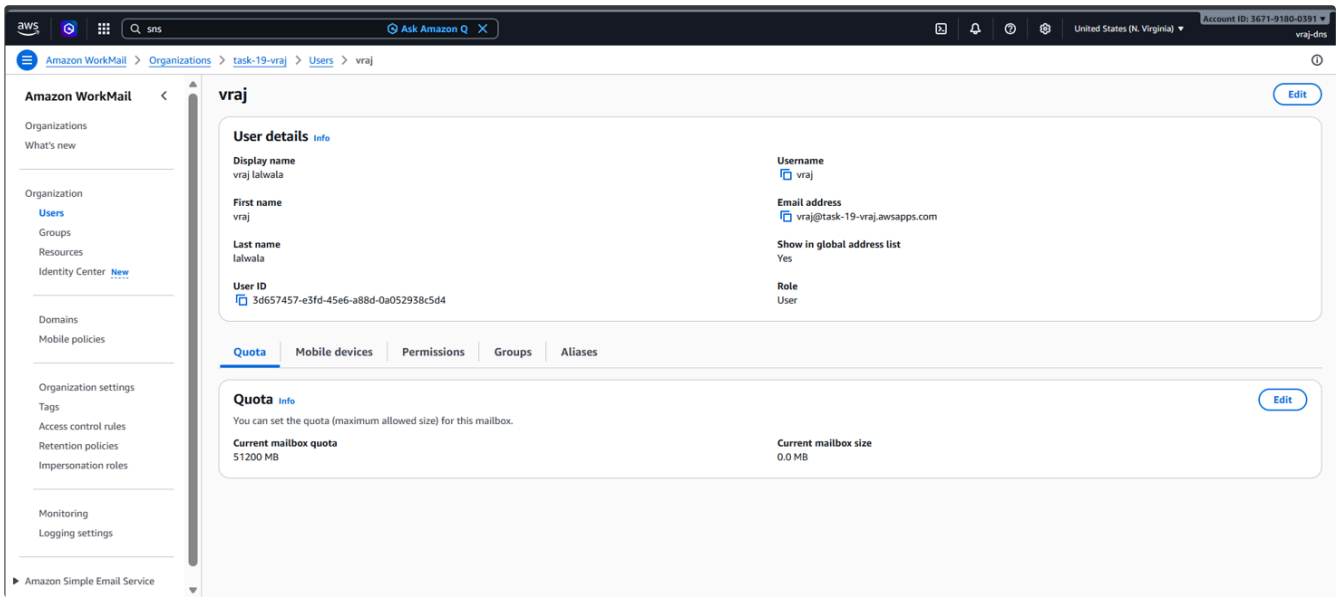

6:05
77%
Test SES > CPU util for SES
1 firing instances
Firing
CPU util for SES
View alert
Summary
Test from SES config from Ubuntu
Values
A=100 C=1
Labels
alertname CPU util for SES
grafana_folder Test SES
instance localhost:9100
Silence
Reply
Forward


Grafana
Alerting › Test SES › Grp-1 › CPU util for SES
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Alert rules
Contact points
Notification policies
Silences
Active notifications
Recently deleted
Settings
Connections
Add new connection
Data sources
Administration
CPU util for SES
Firing
Test from SES config from Ubuntu
Query and conditions
Instances 1
History
Details
Versions
Edit
More
View in Explore
prometheus
Table
{instance="localhost:9100"}
97.1
Threshold
Alert condition
Is above 50
1
Firing

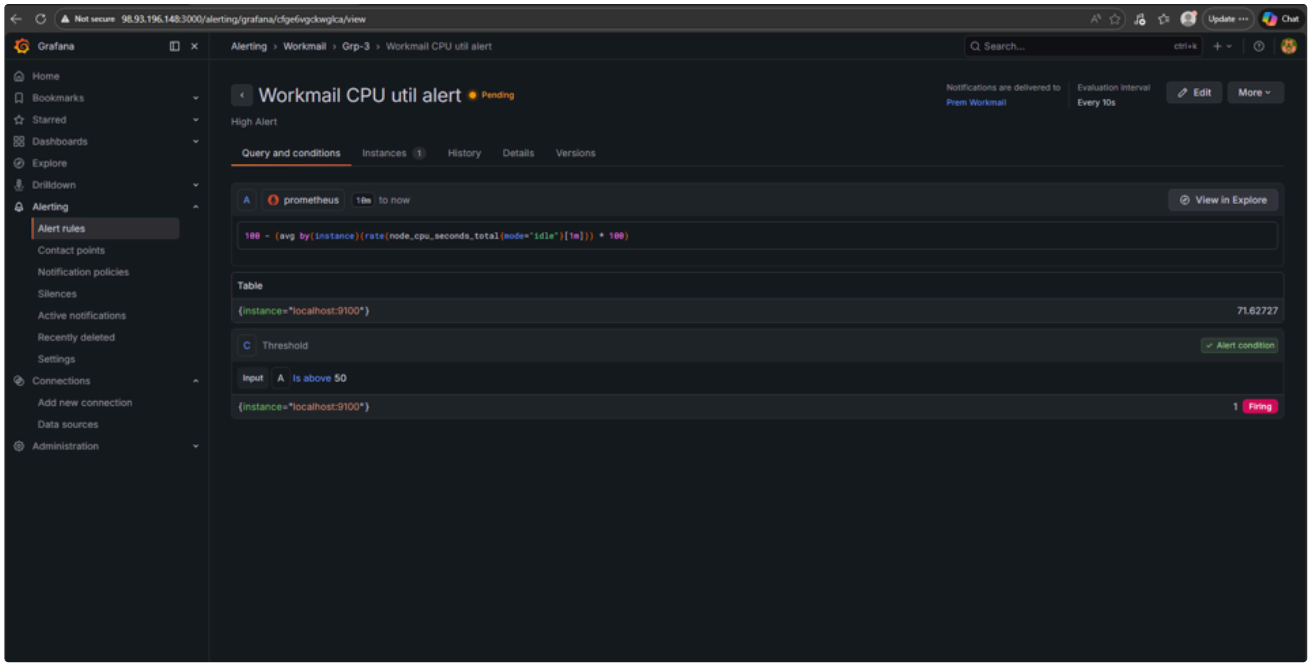
Method 3: AWS WorkMail SMTP

Configuration:

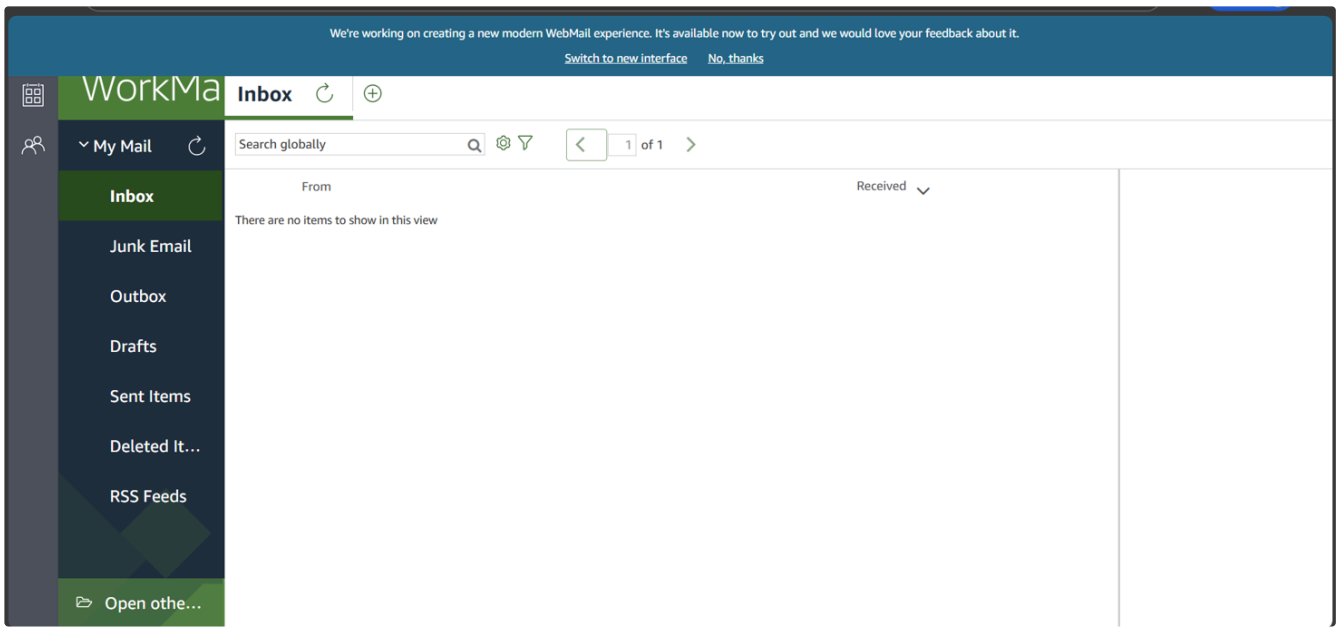
```
#################################### SMTP / Emailing ##########################
[smtp]
enabled = true
host = smtp.gmail.com:587
user = vraj@task-19-vraj.awsapps.com
# If the password contains # or ; you have to wrap it with triple quotes. Ex """#password;"""
password = Vraj@2204
;cert_file =
;key_file =
;skip_verify = false
from_address = vraj@task-19-vraj.awsapps.com
from_name = Grafana Alerts
# EHLO identity in SMTP dialog (defaults to instance_name)
;ehlo_identity = dashboard.example.com
# SMTP startTLS policy (defaults to 'OpportunisticStartTLS')
;startTLS_policy = NoStartTLS
# Enable trace propagation in e-mail headers, using the 'traceparent', 'tracestate' and (optionally) 'baggage' fields (defaults to false)
;enable_tracing = false
```

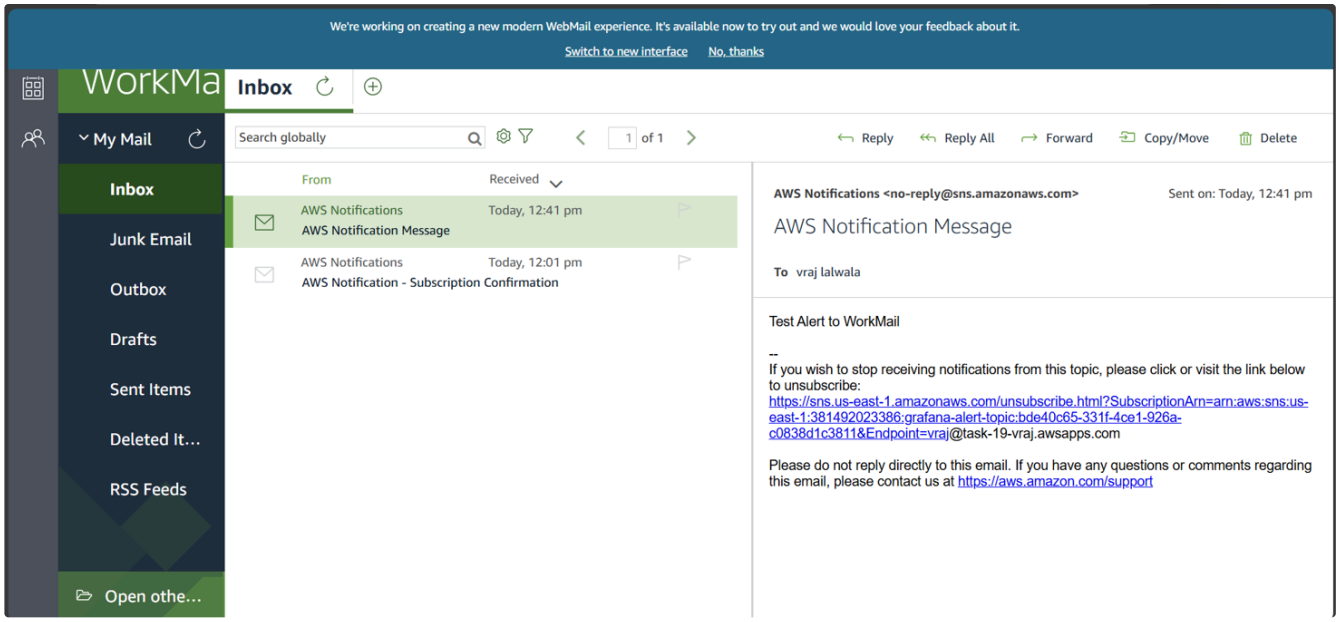


Verification:-


Not secure 98.93.196.148:3000/alerting/grafana/cfge6vgckwgica/view
Grafana
Alerting › Workmail › Grp-3 › Workmail CPU util alert
Search...
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Alert rules
Contact points
Notification policies
Silences
Active notifications
Recently deleted
Settings
Connections
Add new connection
Data sources
Administration
Workmail CPU util alert
Pending
High Alert
Notifications are delivered to
Prem Workmail
Evaluation interval
Every 10s
Edit
More
Query and conditions
Instances 1
History
Details
Versions
A
prometheus
1m to now
View in Explore
100 - (avg by(instance)(rate(node_cpu_seconds_total{mode="idle"}[1m])) * 100)
Table
{instance="localhost:9100"}
71.62727
C
Threshold
Alert condition
Input
A
Is above 50
{instance="localhost:9100"}
1
Firing

Alert Proof:


We're working on creating a new modern WebMail experience. It's available now to try out and we would love your feedback about it.
Switch to new interface
No, thanks
WorkMa
Inbox
My Mail
Search globally
1 of 1
Inbox
Junk Email
Outbox
Drafts
Sent Items
Deleted It...
RSS Feeds
Open othe...
From
Received
There are no items to show in this view



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5. Testing

- CPU alert triggered successfully
- Memory alert triggered successfully
- Alerts delivered via all three methods
- Contact point testing successful

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6. Results

Component	Status
Prometheus	Working
Grafana	Working
Node Exporter	Working
CPU Alert	Working
Memory Alert	Working
Postfix SMTP	Working
AWS SES	Working
AWS WorkMail	Working

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7. Validation

- Alerts generated correctly
- Email notifications received successfully
- All configurations validated

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8. Security Considerations

- SMTP credentials secured
- IAM restrictions applied
- Email access limited to authorized users

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9. Performance Considerations

- Prometheus efficiently scrapes metrics
- Grafana evaluates alerts in real-time
- Email delivery latency minimal

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10. Limitations

- No auto-scaling implemented
- Single-node monitoring setup
- No failover configuration

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11. Future Enhancements

- Integration with Slack/PagerDuty
- High availability Prometheus setup
- Auto-scaling infrastructure
- Centralized logging

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12. Conclusion

The monitoring and alerting system was successfully implemented using Prometheus and Grafana with three email delivery mechanisms. The system ensures reliable alerting through multiple SMTP configurations.

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13. Appendices

- Architecture Diagram
- Use Case Diagram
- Email Alert Screenshots
- Grafana Configuration Screenshots

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14. Customer Acknowledgement

Name: _____

Signature: _____

Date: _____